FRACMATH: numerical damage update

The diagram shows the modified von Mises update. Its equivalent strain is

$$\mathcal{E}_{\mathrm{mvM}}(I_1, J_2) = \frac{k-1}{2k(1-2\nu)} I_1 + \frac{1}{2k}\sqrt{\left(\frac{k-1}{1-2\nu}\right)^2 I_1^2 + \frac{12k}{(1+\nu)^2} J_2}, \qquad k = f_c/f_t.$$

E, ν, f_t, f_c, G_F are Young's modulus, Poisson's ratio, tensile strength, compressive strength and fracture energy. m is the load increment and N the total number of increments. Subscripts e, f, p identify an element, free and prescribed degrees of freedom. $\mathbf{u}$ is nodal displacement, $\mathbf{B}_e$ the strain matrix and $\mathbf{K}_e^0$ undamaged element stiffness. $\mathbf{D}_0$ is the plane-stress matrix form of the elastic tensor. Engineering shear is $\gamma_{xy} = 2\varepsilon_{xy}$. The vectors $\boldsymbol{\varepsilon}_V$ and $\boldsymbol{\sigma}_V$ contain engineering strain and stress components.

$\mathbf{K}^-, \mathbf{K}^+$ are global stiffness before and after the update, $\mathbf{r}^+$ is internal nodal force and η_m the relative free-force imbalance. θ is the crack-normal angle, $\mathbf{n}$ its unit vector, $\widehat{\omega}_e$ trial damage and $\omega_{\max} = 1 - 10^{-12}$ the 2D cap. $\mathbf{D}_{\mathrm{sec}}$ is the secant material matrix. Full definitions are provided in the manuscript and `NOTATION.md`.

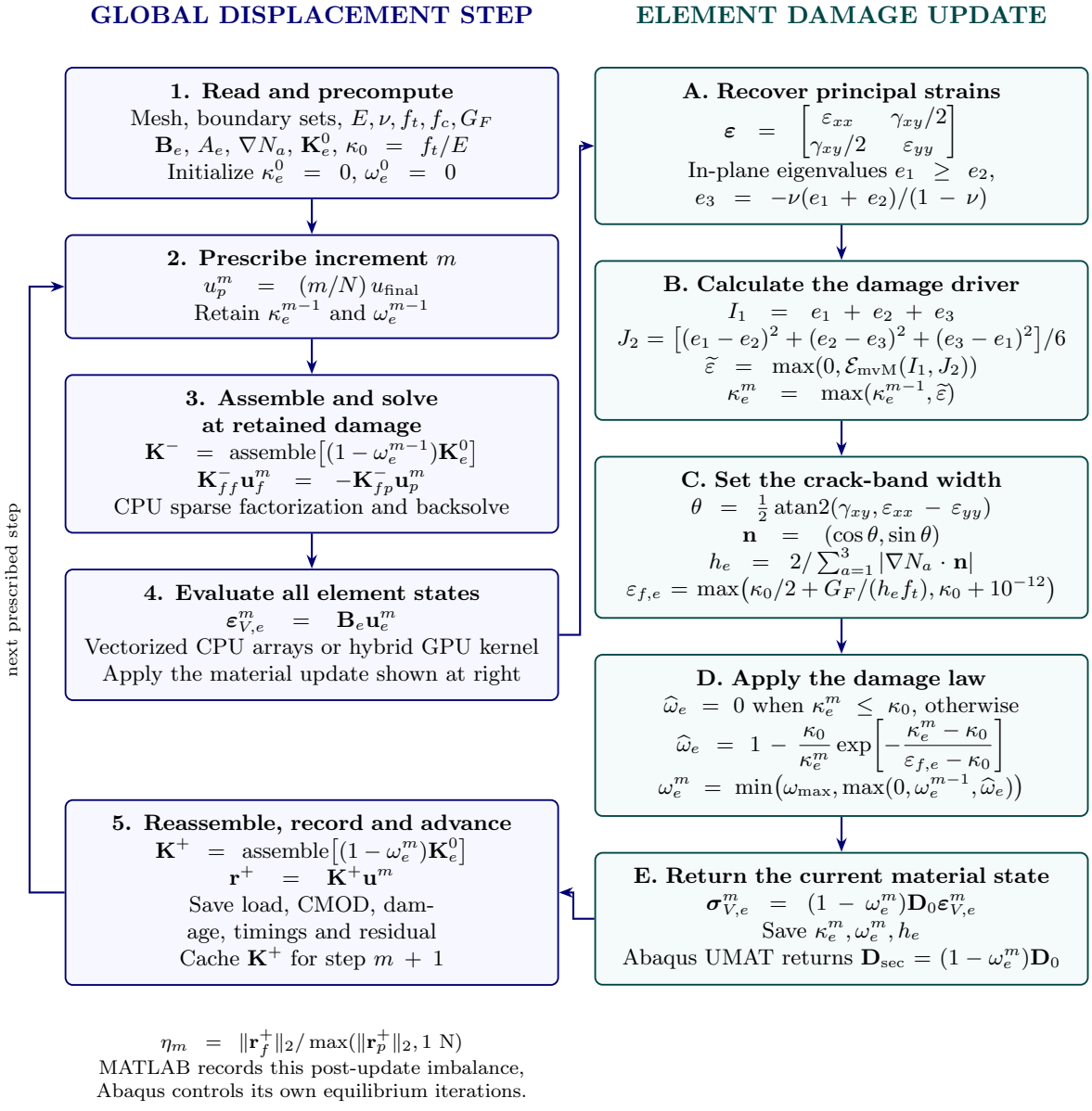


Figure 1: Calculation steps for the 2D bending solver and element damage update. The UMAT uses the same material equations, while Abaqus controls its own equilibrium iterations.